

Muhammad Usman

AI/ML Engineer | Computer Vision, LLMs & RAG Systems | Scalable Backend Engineering

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Summary

AI/ML Engineer with 1.5+ years of production experience shipping **LLM systems, RAG pipelines, and computer vision applications** across defence, enterprise, and SaaS domains. Proven track record of reducing operational overhead, cutting latency, and delivering measurable business impact through intelligent automation. Published in **Springer** and author of an open-source Python package on PyPI. Equally comfortable at the research frontier and in production backend engineering.

Education

Bachelor's Degree in Computer Sciences

Air University, Islamabad, Pakistan

09/2021 – 07/2025

GPA: 3.10 / 4.00

Skills

AI/ML & AI Systems

NLP, PyTorch, TensorFlow, Scikit-learn, OpenCV, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, LLM Fine-Tuning, Semantic Search, AI Agents, Pinecone, FAISS, ChromaDB, OpenSearch, Vector Search, Multi-Agent Systems, Data Pipelines, LiteLLM, Langfuse, Mediapipe

Backend & MLOps/Cloud

FastAPI, Flask, Django, REST APIs, AWS S3, AWS EC2, AWS Lambda, AWS Bedrock, GCP Cloud Run, Docker, Terraform, MLflow, Jenkins, Redis, Prometheus, Grafana, Sentry

Soft Skills

Technical Writing & Documentation, Stakeholder Communication, Analytical Problem-Solving, Cross-Functional Team Collaboration, Research & Experimentation, Agile Workflows

Experience

AI Engineer

Phebsoft

09/2025 – Present

Islamabad, Pakistan

Client Partnership: Gitwork

- Architected **multi-agent systems** capable of autonomously performing tasks such as data scraping, information retrieval, and workflow automation.
- Developed **RAG pipelines with vector embeddings** to enable contextual Q&A across large product documentation datasets.
- Designed scalable AI APIs using **FastAPI**, delivering real-time inference.
- Deployed Docker-based AI microservices on **GCP VM Compute Engine**, streamlining integration between ML models and production backend systems.

AI Engineer

NASTP

02/2025 – 09/2025

Rawalpindi, Pakistan

- Processed and curated **100k+ digital forensic records** for AI model training, fine-tuning, and evaluation.
- Fine-tuned **transformer-based language models** for document classification and semantic retrieval using **AWS SageMaker**, improving categorisation accuracy by **18%**.
- Delivered **5+ RESTful AI APIs** covering OCR, speech, and voice processing using Python and FastAPI.
- Optimised model serving pipelines achieving **<400 ms inference latency** across all production endpoints.

Projects

Afterdesk AI [\[Production Link\]](#)

Stack: OpenCV, Python, FastAPI, PyMuPDF

- Built an AI-powered document automation pipeline using **OpenCV** and computer vision to analyze arbi-

trary PDF templates, detect layout elements and form fields, and automatically map structured data to the appropriate regions.

- Developed a **FastAPI**-based processing service that combines document extraction, template mapping, and automated PDF generation to support dynamic document creation across multiple template formats.

Syscore AI [\[Internal\]](#)

Stack: AWS Bedrock, AWS S3, FastAPI, Python, SYSPRO ERP

- Developed an **AI-powered middleware for SYSPRO ERP** using **AWS Bedrock** for foundation model orchestration and **AWS S3** for XML record storage, enabling natural language and multimodal inputs.
- Achieved **10× faster data retrieval** and **~99% reduction** in ERP request formatting errors by streamlining input parsing and validation.

Speakify AI [\[Production Link\]](#)

Stack: FastAPI, PostgreSQL, Cohere, Pinecone, LiteLLM, Langfuse, Stripe, DocuSign

- Engineered an asynchronous **FastAPI + PostgreSQL** backend across **15+ API domains** for a B2B SaaS marketplace handling the end-to-end speaker booking lifecycle.
- Implemented a **RAG matching engine** using **Cohere embeddings** and **Pinecone** with Langfuse observability; integrated **Stripe** and **DocuSign** for commission splits and contracting.
- Transformed manual workflows into a zero-touch revenue pipeline managing **360+ user profiles** and live bookings.

Kernel-Global [\[Production Link\]](#)

Stack: Python, FastAPI, Azure OpenAI, MySQL, PyMuPDF, python-docx, Docker, Pytest, Locust

- Architected an asynchronous **GenAI microservice** using **FastAPI**, **AsyncIO**, and **Azure OpenAI** to automate candidate assessment, recruitment workflows, and candidate support.
- Built structured document-processing pipelines using **PyMuPDF** and **python-docx** and a conversational AI agent with **MySQL context retrieval**, reducing token usage and API latency by **~35%** through dynamic conversation summarization.
- Implemented **LLM security guardrails** for prompt-injection and jailbreak detection, backed by **40+ unit, integration, and security tests** and Locust-based concurrency testing.

AutoML Platform [\[Internal\]](#)

Stack: Python, MLflow, Docker, FastAPI, Scikit-learn, Prometheus, Grafana

- Delivered a self-service **MLOps platform** automating the model lifecycle across training, evaluation, deployment, and monitoring for multiple ML frameworks.
- Reduced average model deployment time by **~60%** and monitoring setup overhead by **~70%** through automated pipelines and Prometheus/Grafana observability.

View-Invariant Person Re-Identification [\[Research Link\]](#)

Stack: PyTorch, OpenCV, GANs, Attention Mechanisms, Python

- Designed a **Pose-Aware AE-GAN** with an attention-driven feature fusion module to synthesise view-invariant features across non-overlapping camera networks.
- Achieved a **top-1 accuracy gain of 5%+** over baseline models; validated via publication in **The Visual Computer (Springer)**.

Key Achievements

🔗 Publication

View-Invariant Person Re-Identification in Camera Networks by Pose-Aware AE-GAN and Attention-Driven Feature Fusion

Published in **The Visual Computer (Springer)** DOI: doi.org/10.1007/s00371-025-04300-1

📦 Open Source

Conceived and released **ctxintel**, an open-source Python package for LLM context optimisation. Implements a 6-stage deterministic pipeline (ranking, extraction, memory, compression, optimisation) using TF-IDF, spaCy NER, and extractive summarisation zero external API calls.

PyPI: pypi.org/project/ctxintel